

Lab 1

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1 Problem 1

In the first problem, we tried to calculate the period T of oscillation for different initial angular displacements θ_0 . We used runge kutta method to solve the differential equation

$$\frac{d^2\theta}{dt^2} + \omega_0^2 \sin(\theta) = 0 \quad (1)$$

where the equation is split into two first order differential equations:

$$\frac{d\theta}{dt} = p \text{ and } \frac{dp}{dt} = -\omega_0^2 \sin(\theta) \quad (2)$$

The period T is determined by detecting when the pendulum completes a half-period (i.e., it reaches a turning point, where it changes direction). This is when $\theta(t)$ reaches a local maximum.

The time t at this turning point represents half of the period $\frac{T}{2}$. The following figure displays the period T as a function of the initial angular displacement θ_0

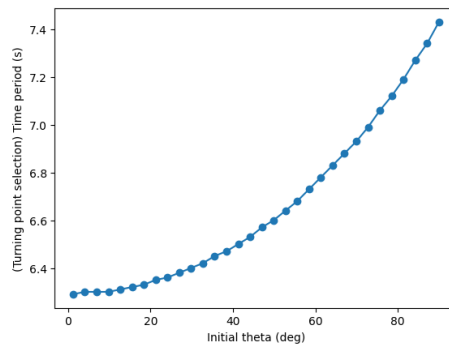


Figure 1

We compared the results from three methods ,small angel approximation, series expansion and integral method to visualize their accuracy and differences

across various initial angles θ_0 . The small angle approximation assumed $\sin(\theta) \approx \theta$ when θ_0 is small. Under this assumption, the period T is constant and given by:

$$T_{\text{harmonic}} = 2\pi$$

. The turning point selection gives an approximation to the period as :

$$T(\theta_0) \approx 2\pi \left(1 + \frac{\theta_0^2}{16} + \frac{11\theta_0^4}{3072} \right)$$

as shown in Figure 2. All methods provided very similar results for small angles, particularly when $\theta_0 < 20^\circ$. Beyond this interval, the power series and integral methods more or less overlap until we reach the angle $\theta_0 = 110^\circ$, where divergence is observed. Thus, all methods provide good results for small angles, while the integral and power series methods are better suited for larger angles.

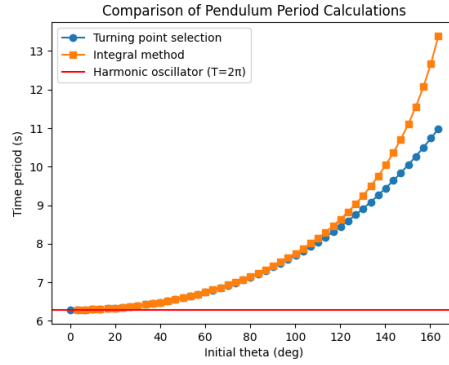


Figure 2

2 Problem 2

Now we consider the problem of having the damped equation:

$$\ddot{\theta} = \gamma\dot{\theta} - \omega_0^2 \sin(\theta). \quad (3)$$

Let $\lambda = 1$, the amplitude $A(t)$ of the damped oscillation decays exponentially over time due to the damping force. This implies that we can model this decay as:

$$A(t) = A_0 e^{-\lambda t}$$

Taking the natural logarithm on both sides gives a linear relationship:

$$\ln A(t) = \ln A_0 - \lambda t$$

where λ is the decay constant. This means that when $\ln A(t)$ is plotted against time t , the result should be a straight line with a slope of $-\lambda$. We can fit a

linear function to the logarithm of the recorded amplitudes:

$$\ln A(t) = a \cdot t + b$$

This is the same as fitting the exponential decay function:

$$A(t) = e^b \cdot e^{a \cdot t}$$

Which is found to be:

$$A(t) \approx 1.65e^{-t/2}$$

To find the characteristic time $\tau = t$ where the amplitude has reduced to half its initial value we do the following:

First, $t_0 = 0 \Rightarrow A(t_0) = 1.65e^0 \Rightarrow A(t_0) = \frac{1.65}{2}$ so we then have :

$$\frac{1.65}{2} = 1.65e^{\frac{\tau}{2}} \Rightarrow \tau = 2 \ln(2).$$

Figure 4 is a plot of the logarithmic swing amplitude vs time.

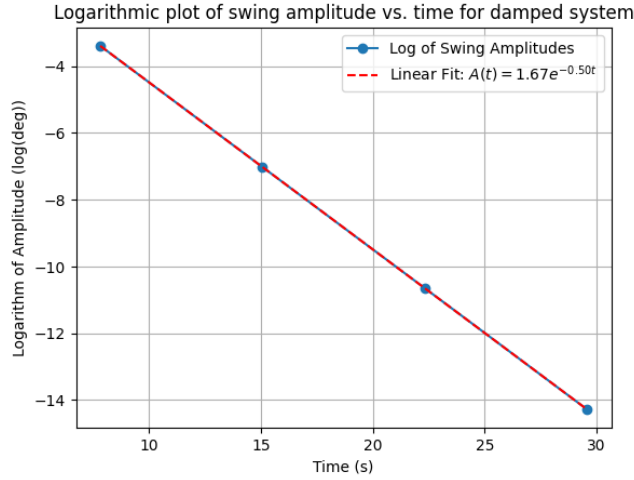


Figure 3: logarithmic swing amplitude vs time

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We explored how the amplitude of the pendulum's first swing is influenced by varying the damping coefficient γ over the range $[0, 1.6]$. For each value of γ , the pendulum was released from an initial angle of $\theta(0) = \frac{\pi}{2}$, and its motion was simulated until the first turning point, where the pendulum changes direction, was reached. The amplitude at this turning point was recorded as a measure of

the pendulum's first swing. The results are shown in the following figure for γ and $1/\gamma$.

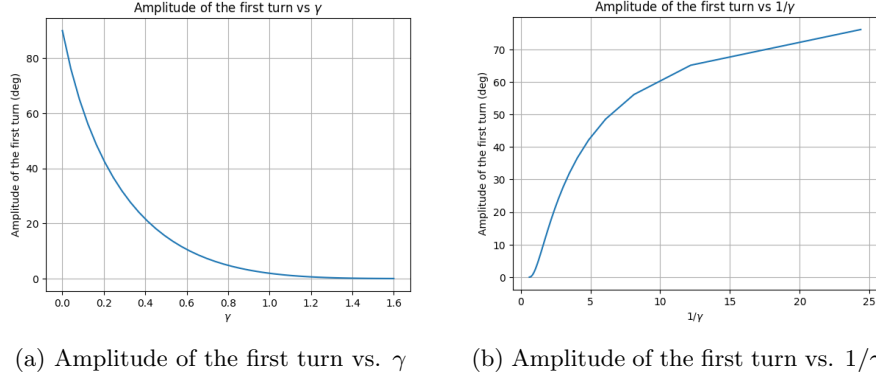


Figure 4: Comparison of the first turn amplitude for varying values of γ and $1/\gamma$.

To answer the question "Can you use one of these plots to extrapolate to the that would give zero amplitude" we see that in plot (a) that for $\gamma > 1.6$ we have a flattening of the plot around 0, so the answer is for $\gamma > 1.6$.

4

Now we work with the damped version of the system represented by

$$\ddot{\theta} + \gamma \dot{\theta} + \omega_0^2 \sin(\theta) = A \cos(\omega t)$$

We ran simulations for different values of the driving force amplitude A , ranging from 0.1 to 0.8 . For each value of A , the pendulum is released from rest with initial conditions $\theta(0) = 0$ and $\dot{\theta}(0) = 0$.

The simulation continues until the system reaches a steady-state oscillation. This steady state is when the amplitude of consecutive oscillations stabilizes, with changes between successive peaks falling below 10^{-6} .

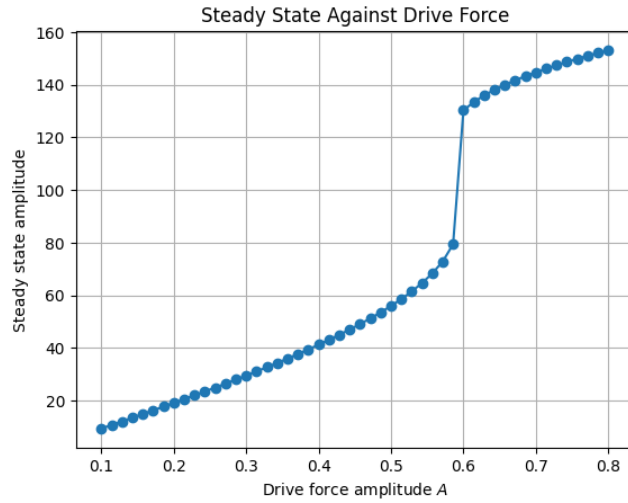


Figure 5

We observed that between $0.1 - 0.6$ we have pretty predictable results for the system, we also see a gradual linear increase of the steady-state amplitude. At 0.6 we have a sharp spike indicating perhaps resonance. After 0.6 we see a stable amplitudes. Now we see how the system behaves when $A = 1$.

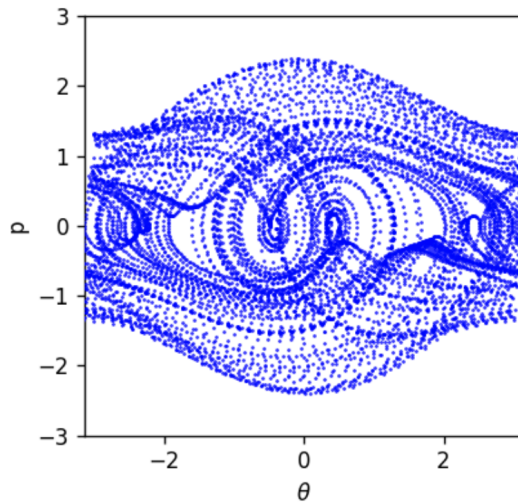


Figure 6: System when $A=1$

The plot shows that the system becomes unstable when $A=1$, perhaps because we have resonance where the driving force frequency matches natural

frequency of the pendulum.